PRACTICAL 1

Aim: Study of software and hardware requirements of various Operating Systems (Windows/Linux/Unix) and Implementation of DOS commands on command prompt.

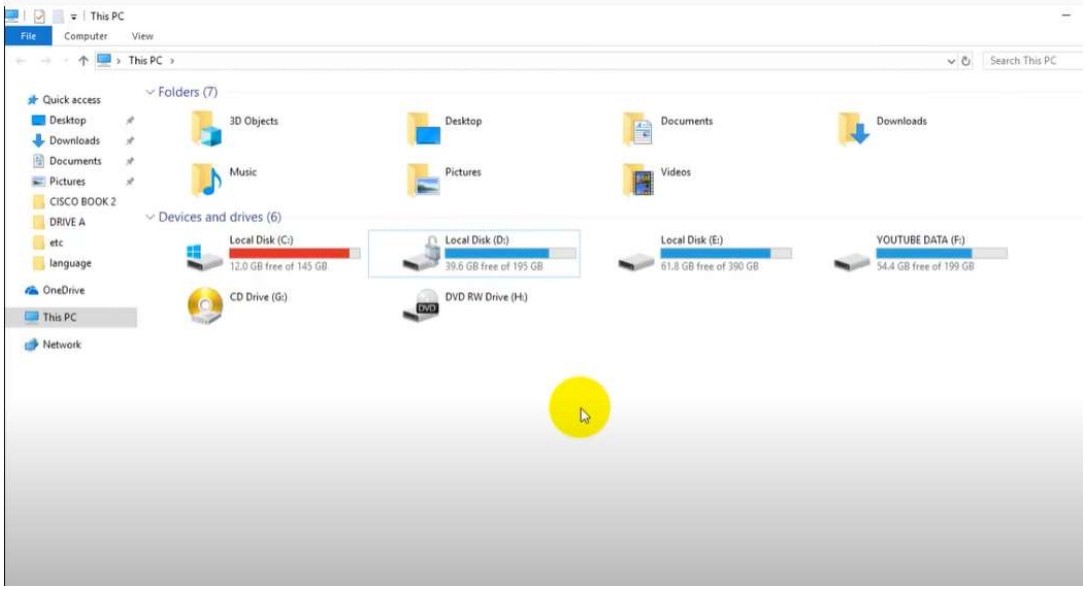
A. Windows

Here's what it takes to upgrade to Windows 10 on your PC or tablet:

- **Processor:** 1 gigahertz (GHz) or faster processor or SoC
- **RAM:** 1 gigabyte (GB) for 32-bit or 2 GB for 64-bit
- **Hard disk space:** 16 GB for 32-bit OS or 20 GB for 64-bit OS
- **Graphics card:** DirectX 9 or later with WDDM 1.0 driver
- **Display:** 800 x 600

Below are some additional details regarding requirements for key features:

1. **BitLocker Drive Encryption**
   - BitLocker Drive Encryption is a data protection feature that integrates with the operating system and addresses the threats of data theft or exposure from lost, stolen, or inappropriately decommissioned computers.



   - (available with Windows 10 Pro or Windows 10 Enterprise only) requires a **Trusted Platform Module (TPM) 1.2** or higher and Trusted Computing Group (TCG)-compliant BIOS or UEFI.
   - BitLocker can be used on devices without TPM, but you will need to save a startup key on a removable device such as a USB flash drive.
   - TPM 2.0 and InstantGo support is required when you want to automatically encrypt the local drive when joining a device to Azure Active Directory (AAD). Check with your PC manufacturer to confirm if your device supports the correct TPM version and InstantGo for the scenario you want to enable.

2. BitLocker To Go is BitLocker Drive Encryption on removable data drives. This feature includes the encryption of:

- USB flash drives

- SD cards
- External hard disk drives
- Other drives that are formatted by using the NTFS, FAT16, FAT32, or exFAT file system.

As with BitLocker, you can open drives that are encrypted by **BitLocker To Go** by using a password or smart card on another computer.

In Control Panel, use **BitLocker Drive Encryption**. It requires a USB flash drive (available in Windows 10 Pro and Windows 10 Enterprise only).

3. **Client Hyper-V**
   - Many of you need to run multiple operating systems.
   - Hyper-V lets you run multiple operating systems as virtual machines on Windows.
   - Hyper-V specifically provides hardware virtualization.
   - That means each virtual machine runs on virtual hardware.
   - Hyper-V lets you create virtual hard drives, virtual switches, and a number of other virtual devices all of which can be added to virtual machines.
   - It requires a 64-bit system with second level address translation (SLAT) capabilities and additional 2 GB of RAM (available in Windows 10 Pro and Windows 10 Enterprise only).
4. **Cortana**
   - It is a virtual assistant developed by Microsoft which uses the Bing search engine to perform tasks such as setting reminders and answering questions for the user.
   - It is only currently available on Windows 10 for the United States, United Kingdom, China, France, Italy, Germany, Brazil, Mexico, Japan, Canada, Spain, Australia and India.
5. **Microsoft account** is required for some features.
6. **Miracast**
   - Miracast is a wireless display standard designed for mirroring a smartphone, tablet, or PC's screen to a television without requiring any physical HDMI cables. It's becoming more widespread with each passing day.
   - Itrequires a display adapter which supports Windows Display Driver Model (WDDM) 1.3, and a Wi-Fi adapter that supports Wi-Fi Direct.
7. **Movies & TV** application is not available in all regions. For the most up-to-date list of regions, please go to the **Movies & TV** information page.
8. **Secure boot** requires firmware that supports UEFI v2.3.1 Errata B and has the Microsoft Windows Certification Authority in the UEFI signature database.
9. **Skype** is available only in select countries and regions. Calling to select countries and regions only. Excludes special, premium and non-geographic numbers. For details, visit the **Office FAQ page.**
10. **Snap**: The number of applications that can be snapped will depend upon the minimum resolution for the application with a limit of two applications in Tablet mode and four applications in Desktop mode.
11. **Speech recognition**

It will vary by device microphone. For a better speech experience, you will need a:

i. High fidelity microphone array
ii. Hardware driver with microphone array geometry exposed

12. **Tablet mode** is available on tablets and 2-in-1s with GPIO indicators or those that have a laptop and slate indicator will be able to be configured to enter "tablet mode" automatically.
13. **Touch**: To use touch, you need a tablet or a monitor that supports multi-touch.
14. **Two-factor authentication** requires the use of a PIN, Biometric (finger print reader or illuminated infrared camera), or a phone with Wi-Fi or Bluetooth capabilities.
15. **Windows Hello** requires a camera configured for near infrared (IR) imaging or fingerprint reader for biometric authentication. Devices without biometric sensors can use Windows Hello with a PIN or a portable Microsoft compatible security key.
16. **Xbox**

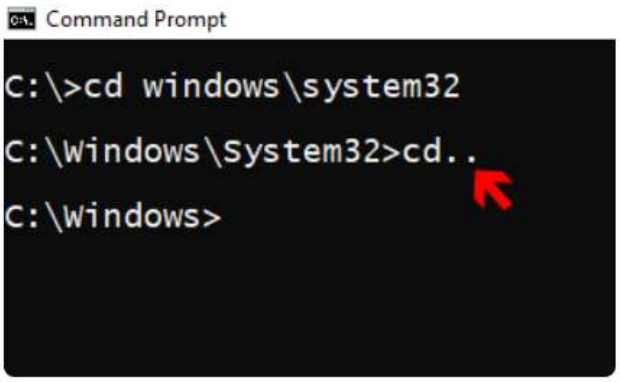
- It is a video gaming brand created and owned by Microsoft. The brand consists of five video game consoles, as well as applications (games), streaming services, an online service by the name of Xbox network, and the development arm by the name of Xbox Game Studios. The brand was first introduced in the United States in November 2001, with the launch of the original Xbox console.
- Itsapplication requires an Xbox Live account, which is not available in all regions.

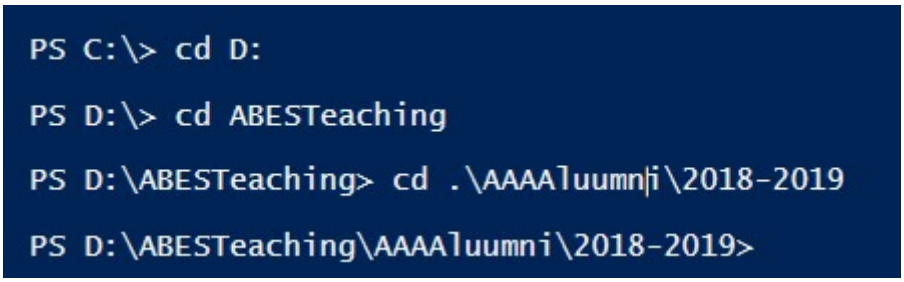
17. **Wi-Fi Direct Printing** requires a Wi-Fi adapter that supports Wi-Fi Direct and a device that supports Wi-Fi Direct Printing.

Windows Command

Part I: Important commands

1. **driverquery**
   Drivers are very important in your PC. Missing a important driver can hamper your work. Use driverquery command to get a full list of installed drivers in your pc. It'll help you to find the missing driver.
2. **ipconfig**
   ipconfig will provide you your ip address along with your local network.
   ipconfig:
   Displays the current IP address of your computer and the DNS server address. Try and observe the outputs of "ipconfig" and "ipconfig /all".
   What are the IP and MAC (Media Access Control) addresses of your computer?
3. **systeminfo**
   Use systeminfo to know very basic information about your pc's hardware, like – motherboard, processor & ram.
4. **ping**
   The ping command sends packets of data to a specific IP address (or domain) on a network and then lets you know how long it took to transmit that data and get a response. If you get the response properly then the connection of the device is working properly if not a particular server or your online connection is blocking communication between yourcomputer and another.
   ping <ip or domain>
5. **tasklist**
   Use tasklist to get current list of all tasks running on your pc.
6. **Change the drive**
   Use the cd.. command to go one folder up.





7. **Create a new directory / Folder (mkdir)**

   You can make a new folder using the mkdir (Make Directory) command. The syntax of these commands is mkdir Folder.

8. **Dir**
   To test if it worked, use the dir command. The newly created folder appears in the list.

   If you are working on the "C:" drive and you want to create a new folder in "D:," called Google, type mkdir d:\Google and then press Enter.

Type "dir" to see the list of the files and "dir /a" to see the list of the hidden files as well. Make hello.txt visible again.

9. **Cls**
   To clear the existing commands in prompts type cls and press Enter.cls

Part II. In this part basic DOS commands will be introduced.

1. Execute the two MS-DOS commands which will set the default to the root directory (\) on the d: drive.
d:
cd \

2. Type **date** and **time** and then press ENTER. DOS will let you check and change the date and time. If the correct date/time is displayed, simply press ENTER. If the date/time is incorrect, type the correct date/time and press ENTER.

3. Type **ver** to display the Windows OS's version number.

4. Use the**dir** command to list the contents of the root directory on the d: drive (i.e. d:\>dir).

5. Type "**help** [command]" to display help information on that command. (i.e. help dir)

Part III. In the steps below we will create a simple text file called hello.txt, make two directories, mydir and mysubdir and copy hello.txt into the subdirectories. We will then delete the copies of hello.txt and remove the directories under mydir.

1. Invoke the MS-DOS full screen text editor utility EDIT (if your OS does not support EDIT then use notepad command i.e. notepad hello.txt) to create a text file called hello.txt:
d:\> notepad hello.txt
Enter the line “Hello World!” use File|Save and File|Exit to save your work and to quit EDIT.

2. Use the type command to display contents of hello.txt.
d:\> type hello.txt
You should see: Hello World!

3. Use the md command to create (make) a new directory called mydir. d:\> md mydir

4. Use the cd command to change the default directory to mydir. d:\> cd mydir

5. Use dir to display the contents of mydir. It should contain no files except for "pointers" to itself and its parent.
d:\mydir>dir

6. Use the copy command to copy the file hello.txt in the root directory to mydir.
d:\mydir> copy d:\hello.txt

7. Use dir to display the contents of mydir. You should see hello.txt listed.
d:\mydir>dir

8. Use md to create (make) a subdirectory for mydir called mysubdir.
d:\mydir> md mysubdir

9. Use cd to change the default directory to mysubdir. d:\mydir> cd mysubdir

10.Copy hello.txt to mysubdir. Use the mydir directory copy of hello.txt. d:\mydir\mysubdir> copy d:\mydir\hello.txt

11. Make a second copy of hello.txt but call it hello1.txt. d:\mydir\mysubdir> copy d:\hello.txt hello1.txt

12. Use the ren (rename) command to rename hello.txt as hello2.txt. d:\mydir\mysubdir> rename hello.txt hello2.txt

13.Check your work - display the contents of mydir. d:\mydir\mysubdir> tree \mydir

14.Check that the content of hello2.txt has not changed by displaying it.
d:\mydir\mysubdir> type hello2.txt

15. Using a wildcard delete all files with a .txt extension in mysubdir.
d:\mydir\mysubdir>del *.txt

16. Use the dir command to check that both files are gone. d:\mydir\mysubdir>dir

17.Return to the mydir directory. d:\mydir\mysubdir> cd \mydir

18. Use the dir command to view the contents of the mydir directory.
d:\mydir>dir

19. Use the rd command to remove the mysubdir directory.
d:\mydir>rdmysubdir

20. Use the dir command to check that the mysubdir subdirectory is gone. d:\mydir>dir

21. Return to the root directory. d:\mydir> cd \

Part IV: Basis commands and their functions
cd : Change directory or display current directory path.
cls : Clear the window.
dir : Display list of contents of current directory.
help : Display list of commands or help about a command.
notepad : Run the Windows Notepad text editor.
type : Displays the contents of a text file.
attrib : Displays or changes file attributes.
assoc : Displays or modifies file extension associations.
chkdsk: Checks a disk and displays a status report.
color : Sets the text and background colors.
comp : Compares the contents of two files or sets of files.
copy : Copies one or more files to another location.
date : Displays or sets the computer's date
del(or erase): Deletes one or more files.
echo : Displays messages, or turns command echoing on/off.
exit : Closes the DOS window.
find : Searches for a text string in a file or files.
md (or mkdir) : Creates a directory.
more : Displays the contents of a file one screen at a time.
move : Moves one or more files from one directory to another directory.
rd (or rmdir) : Removes a directory.
ren (or rename) : Renames a file or files.
sort : Sorts input.
time : Displays or sets the computer's time.
tree : Graphically displays the directory structure of a drive or directory.
xcopy : Copies files and directory trees.

B. **LINUX:-** Write hardware requirement, software requirement and 5 commands and run it on any online Linux virtual environment

C. **UNIX:-** Write hardware requirement, software requirement and 5 commands and run it on any online UNIX virtual environment

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